Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/03/2004
PI Name:	Drysdale, Alan E Ph.D.		
Project Title:	Equivalent System Mass Impacts of Clothing on Wastes and Mission Cost		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32815	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	03/03/2003	**End Date:**	03/02/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Garton, Harry (Boeing) Hasselbrack, Sally (Boeing) Barker, Roger (NCSU) Thompson, Don (NCSU) Scruggs, Barbara (NCSU) Fratto, Lisa (NCSU)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Resupply of clothing comprises about 16% - 19% of the mission ESM for long duration missions. Disposing of the clothing as waste also constitutes a significant impact. Two approaches have been identified that show promise for large reductions in these impacts. · Mass of clothes can be reduced by using high-tech materials, which also provide other comfort and safety benefits. · Mass of clothes can be reduced by using a cleaning system and reusing clothes. This does require additional equipment and interfaces, but preliminary analyses show promise that the net ESM would be greatly reduced. For the International Space Station (ISS), crew clothing is supplied at a rate of about 194 kg and 1 m3 per 90-day increment for a crew of 3, though an attempt to reduce this is underway. The clothing, once worn, then becomes waste that must be managed. Crew debriefs (Increment 1) documented that odors produced from soiled clothing were more unpleasant for the crew than were odors from food trash. We propose to evaluate standards and requirements for providing clothing for human spaceflight missions, using ISS as a baseline and modifying and expanding the requirements for future missions. We will address two possible ways to reduce the mission impact of clothing; a redesign of the wardrobe, and cleaning and re-use of clothing in space. We will identify life-support related impacts that clothing has on subsystems and interfaces, options for processing the clothing, costs and benefits, and recommend particular approaches for particular near-term missions. Reduce clothing system ESM. The clothing system is one of the few areas where significant improvements in the ALS project metric can be made.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential cross utilization of developed technology. In particular, reduced water and energy use, and lower vibration and noise, are likely.
Task Progress:	Identified need for clothing in space. Identified applicable NASA and commercial standards. Identified interfaces with other systems. Identified current (STS and ISS) and historical approaches and ISS use rates. Began identifying differences between past and current missions and ALS missions, and the consequent differences in clothing system requirements. Current clothing systems have been assessed and a preliminary set of requirements for advanced missions developed. Assessed fire hazards and need for / practicality of fire retardant clothing. ESM for the ISS baseline approach has been calculated. Clothing use is about 10 kg per person plus 0.625 kg per person per day. ESM for using a terrestrial COTS washer has been calculated, to provide a baseline for clothes washing. Water regeneration is a critical issue with regard to clothes washing. Water regeneration costs for a range of NASA and terrestrial technologies were calculated. Commercial clothing standards have been identified, mostly for protective clothing. Began assessment of cleaning options.
Bibliography Type:	Description: (Last Updated: 04/14/2021)
Papers from Meeting Proceedings	Drysdale A, Garton H, Barker RL, Thompson DB, Scruggs BJ, Fratto L. "Clothing systems for long-duration space missions." International Conference on Environmental Systems, Colorado Springs, CO, July 2004. International Conference on Environmental Systems, Colorado Springs, CO, July 2004. SAE Technical Paper 2004-01-2580. https://doi.org/10.4271/2004-01-2580 , Jul-2004